Aayush Kumar Chaudhary

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Research Interests: Image processing, computer vision, eye-tracking, machine learning (deep learning), Human Computer Interaction

EDUCATION

Ph.D. student, Chester F. Carlson Center for Imaging Science — **2016 – Present**
Rochester Institute of Technology — Rochester, New York
Area of focus: eye tracking methodology (image processing and machine learning)
Relevant courses: image processing and computer vision, deep learning for vision, AI explorations, multiview geometry, human visual system, Bayesian statistics for machine learning

Bachelor of Engineering (B.E.), Electronics and Communication Engineering, Pulchowk Campus, I.O.E., Tribhuvan University — **2010 – 2014**
Kathmandu, Nepal
(University Rank: 1st 85.47%) [Engineering Stream]
Thesis: Teleoperation of Mobile Robotic Arm Through Imitation of Human Arms Movement

SKILLS

Programming languages	**Python,** Matlab, C, C++
Packages	**PyTorch, TensorFlow, Scikit-Learn, OpenCV,** VisualSFM
Eye trackers	Pupil Labs eye trackers, custom-made remote eye trackers with mirrorless cameras
Miscellaneous	Human subjects research, robotics, circuit design tools (Kicad, Proteus), photography (iris)

RESEARCH EXPERIENCE

Multidisciplinary Vision Research Laboratory, RIT, Research Assistant
Advisor: Dr. Jeff Pelz. — **June 2017 – Present**

Gaze Estimation
- Study of feasibility of iris features based gaze estimation
- Developed a new approach of incorporating regression-based models with machine learning based approaches
- Improved the precision and accuracy of eye trackers
- Designed and tested eye tracking hardware setup for study of various eye movements for human subjects

Segmentation of eye-regions
- Improved the robustness of eye-parts segmentation with memory efficient models.
- Designed novel framework for ellipse segmentation on eye-parts
- Currently exploring temporal segmentation

Privacy in eye videos
- Designed concept for a novel approach to preserve privacy in the eye videos without degrading the accuracy of eye trackers

Synthetic dataset for eye-tracking applications
- Collaborated with multiple research labs in RIT to introduce a synthetic eye image generation platform for eye-tracking applications (https://cs.rit.edu/~cgaplab/RIT-Eyes/)

PUBLICATIONS

JOURNAL ARTICLES

[J.2]' *[arXiV] [Currently in review]*
Chaudhary, A. K. & Pelz, J.
π_t:Enhancing the Precision of Eye Tracking using Iris Feature Motion Vectors

[J.1] ***[JEMR 2019]***
Journal of Eye Movement Research
Chaudhary, A. K. & Pelz, J.
Motion tracking of iris features to detect small eye movements.

CONFERENCE PROCEEDINGS

[C.6]' *[arXiV] [Currently in review]*
(equal contribution)*
Kothari, R.*, **Chaudhary, A. K.*,** Bailey, R. J., Pelz, J., Diaz, G.
EllSeg: An Ellipse Segmentation Framework for Robust Gaze Tracking

[C.5] ***[ACM-SAP 2020]***
Symposium of Applied Perception
Nair, N., Kothari, R., **Chaudhary, A. K.,..,** Bailey, R. J.
RIT-Eyes: Rendering of near-eye images for eye-tracking applications
[Best Paper Honorable Mention] *(oral)*

[C.4] ***[ACM-ETRA 2020]*** *(short-paper)*
Eye Tracking Research & Applications
Chaudhary, A. K. & Jeff B. Pelz. 2020.
Privacy-Preserving Eye Videos using Rubber Sheet Model applications

[C.3] ***[ACM-ETRA 2020]*** *(adjunct)*
Eye Tracking Research & Applications
Nair, N., **Chaudhary, A. K.,** Kothari, R., Diaz, G., Pelz, J., Bailey, R.
RIT-Eyes, realistically rendered eye images for eyetracking applications

[C.2] ***[ACM-ETRA 2019]***
Eye Tracking Research & Applications
Chaudhary, A. K.
Motion tracking of iris features for eye tracking.

OPEN CHALLENGES

[ICCVW-2019]
IEEE/CVF International Conference on Computer Vision Workshop
Chaudhary, A. K.*, Kothari, R.*, Acharya., M.*, Dangi., S,.., Pelz., J.
RITnet: Real-time Semantic Segmentation of the Eye for Gaze Tracking
[Semantic Segmentation Challenge: Winner] *(oral)*

WORK EXPERIENCE

Chester F. Carlson Center for Imaging Science, RIT — **Rochester, New York**
Teaching Assistant (Undergraduate Courses Assisted: Vision and Psychophysics, Linear Methods of Imaging) — Aug 2016-May 2017

Samsung India Electronics Pvt. Ltd (SIEL, Nepal branch) — **Kathmandu, Nepal**
Senior Engineer (Smart phones performance testing and validation (METI team)) — Nov 2014-July 2016

- My contribution in Q1'2016 was recognized with a performance award

SCHOLARSHIPS, AND AWARDS

Professional Achievement

2016 **Performance Award in recognition of High Performance for Q1'2016**, Samsung (SIEL)

Academic Achievement

2019 **Prof. F.N.Trofimenkoff academic achievement award,** a prestigious award given to one student each year by Department of Electronics and Computer Engineering, Pulchowk Campus, IOE (DoECE) for academic success during undergraduate studies.

2019 **Winner of Facebook OpenEDS competition.** Our model is memory efficient and achieves the state-of-the-art results on the 2019 OpenEDS Semantic Segmentation challenge.

2015 **NCell Excellence Award,** an award for being the overall topper of DoECE

2010/14 **University Rank 1st (Engineering Stream),** Rank First in All Bachelor's Degree programs of I.O.E. Tribhuvan University.

2014 **NCell Scholarship Award,** Third-year topper of DoECE

2010 **Nepal Board Topper,** Scored the highest score in the country in Higher Secondary Education Board (12th grade) Science faculty.

Merit Scholarship

2016- **RIT Ph.D. Scholarship/Assistantship,** Financial Support for Ph.D. studies at RIT.

2010/14 **College Fellowship,** Awarded for outstanding academic performance in each semester of Bachelor of Engineering by I.O.E., Pulchowk Campus.

2010/14 **Full Academic Scholarship, a**warded to pursue B.E. at I.O.E, Pulchowk Campus based on nationwide competitive examination.

Activities and Leadership

2014 **Coordinator, LOCUS 2014 (Technological Event) -** Coordinated and organized the 11th national technological festival- the biggest technological festival of the country for DoECE and Department of Electrical Engineering**.**

2013/14 **Secretary, Amateur Radio Club,** A licensed Ham-Radio Operator

2013 **Team Member, ABU Robocon 2013, Vietnam -** Integral part of the hardware team (designing and building circuit boards) for the both manual and automatic robot that represented Nepal in the Asia-Pacific Broadcasting Union (ABU) Robocon 2013, Vietnam.

2013 **Secretary, Electronics Club -** Organized crash course to freshmen regarding Electronics circuits and helped to launch the first magazine of Electronics Club 'Graphene'.

2018 **IdeaLab -** Presented TUBBLE, an engaging Toy for Active Kids.

2014 **Child App Competition -** Presented a social interacting app 'Beautiful Minds' for autistic children and was one of the finalists of the Child App Competition. It was organized by Microsoft Innovation Center (Nepal) and UNICEF Nepal.

Miscellaneous

[Poster.3] RITnet: Real-time Semantic Segmentation of the Eye for Gaze tracking
- ICCV 2019, Workshop on Eye Tracking for VR and AR

[Poster.2] Motion Tracking of Iris Features for Eye Tracking.
- In 2019 Symposium on Eye Tracking Research and Applications (ETRA'19)

[Poster.1] Motion Tracking of Iris Features to Detect Small Eye Movements.
- Graduate Showcase, RIT, 2018

[Talk.2] RIT-Eyes: Rendering of near-eye images for eye-tracking applications
- Talk, in ACM-SAP 2020 (virtual)

[Talk.1] RITnet: Real-time Semantic Segmentation of the Eye for Gaze tracking.
- Talk, in ICCV 2019, Workshop on Eye Tracking for VR and AR.

[Reviewer.4] Pattern Analysis and Applications (PAAA)

[Reviewer.3] Imaging Science Journal (IMS)

[Reviewer.2] Eye Tracking Research and Applications (ETRA)

[Reviewer.1] Journal of Eye Movement Research (JEMR)

[Judge.1] Judge, LOCUS 2016 (Technological Event)

- Represented SIEL, Nepal branch as one of the main judges of the event Samsung Galaxy App Quest conducted to promote application development in Nepal.

[Mentor] High school students (2019, 2020)

Actively mentoring and working closely with numerous undergraduate and graduate level students